A project report on

GENERATE ELECTRICITY BY WASTE MATERIAL

Submitted in the partial fulfillment of the requirements for the award of degree

DIPLOMA

In

ELECTRONICS AND COMMUNICATION ENGINEERING

Under the esteemed guidance of

M.PAVANI, B.Tech

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Department Of Electronics & Communication Engineering

BAPATLA POLYTECHNIC COLLEGE::BAPATLA

BAPATLA-522101, A.P

2021-2024

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

BAPATLA POLYTECHNIC COLLEGE::BAPATLA

CERTIFICATE

This is to certify that the project report entitled "GENERATE ELECTRICITY BY WASTE MATERIAL" is a bonafide record work done by Mr/Ms. B. Janaki Ramana Regd.No 21106-EC-041, under the guidance and supervision of Mr. **M.PAVANI, B.Tech** submitted in the partial fulfillment of the requirements for the award of the degree Diploma in Electronics and Communication Engineering by S.B.T.E.T, Vijayawada during the academic year 2021 – 2024.

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ACKNOWLEDGEMENT

The successful completion of any task is not possible without proper Guidance, suggestions and environment. The combination of all these factors act as a backbone to our project titled "GENERATE ELECTRICITY BY WASTE MATERIAL".

We express our great pleasure to our honorable principal Mr. **T.RAJENDRA PRASAD, M.Tech,** who inspired us a lot through his valuable messages. He is only the personality who had given the real meaning to the technology studies.

We express our sincere and heart full thanks to our H.O.D. Mr. **G.SUBBA RAO, B.Tech,** for his motivation, endeavor and also his constant help & support throughout our project.

We express our whole hearted thanks to our beloved guide Ms. **M.PAVANI, B.Tech,** for his splendid effort and guidance throughout the course of our project work, without whom the assignment would not have been successfully completed.

Our Special thanks to all the teaching and non-teaching staff of the department for their cooperation in completing the project successfully.

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DECLARATION

We the students of BAPATLA POLYTECHNIC COLLEGE, BAPATLA hereby declare this project work entitled "GENERATE ELECTRICITY BY WASTE MATERIAL" is being submitted to the department of E.C.E, for the award of Diploma degree in Electronics and Communication Engineering, is a bonafide record work done by us at the college and it has not been submitted to any other Institutes for the award of any other degree.

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GENERATE ELECTRICITY BY WASTE MATERIAL

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ABSTRACT

In This Project We use waste materials for generate Electricity We show in this project one Electricity generating zaar box when we have waste materials like plastic, paper and other Then we burn that Materials in zaar box and when burning start then heat going to heating penal then heating penal convert the heat into Electricity Then we store that Electricity in battery and use that Electricity for bulb glowing and many others work

This is Live working idea for generate Electricity by Plastic and Waste Materials , In This Project when electricity start storing that time output power supply off because we use heating sensor so when electricity store perfect then heating sensor turn on the output power supply and LED bulb start glowing and we can show that time live working of generate electricity by waste materials

CHAPTER 1
INTRODUCTION

Think Placements . . Think NEC

Introduction:

The Purpose of making this project is to generate electrical energy from bad materials like plastic, rubber, garbage and bad stuff etc. and store that electrical energy in the battery through the circuit and use that electrical energy to operate the whole project. And the LED bulb is shown to be turned on

In This Project when burning start then heating generate and heating penal start converting heat to electricity and that electricity we can see on multi meter display , we can see how much voltage generate by waste materials and we electricity generating perfectly then automatic heating sensor on the output power supply then Big LED Bub start glowing and our idea everyone can see in live working , Our Idea 100% work for generate electricity by waste materials . So this is our best live working idea

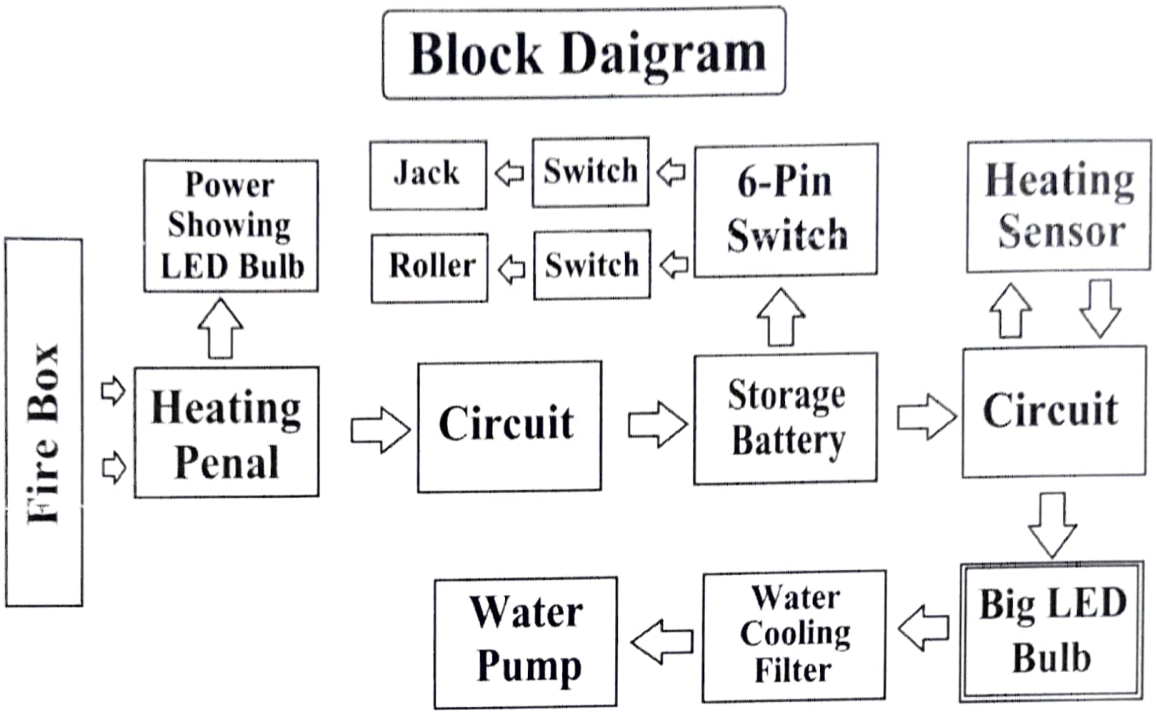
Filter Pollution Control Filter - In This Idea we show theoretically Idea Based Pollution Control System , In This Project When Smoke generate then smoke go to by pipe line to water tank then on water top surface corban start collecting ,and water can not heat so there we used water colling filter that filter cool the water again and again and in this idea only we control corban pollution so we collect on water tank up side corban by smoke .

Methodology:

The first step before the project implementation was to review The project scope and research area. Then the next task was to Design the mechanical structure and electrical structure of the conveyor belt which is To be built. Then, if all the design had been finalized, the implementations of the hardware and the circuitry took place. Reaching the pick of the project, the programming segment Took place especially for the heating penal output , heating sensor sensing process and Output to the LED Bulb glow for . Last But not least, certain modification on the circuitry and soft-Ware took place in order to make the system perform in finer Movements. Thus, troubleshooting process also took place to Correct certain faulty processes while the system was performing its task

Think Placements

Block Diagram:



Literature Review:

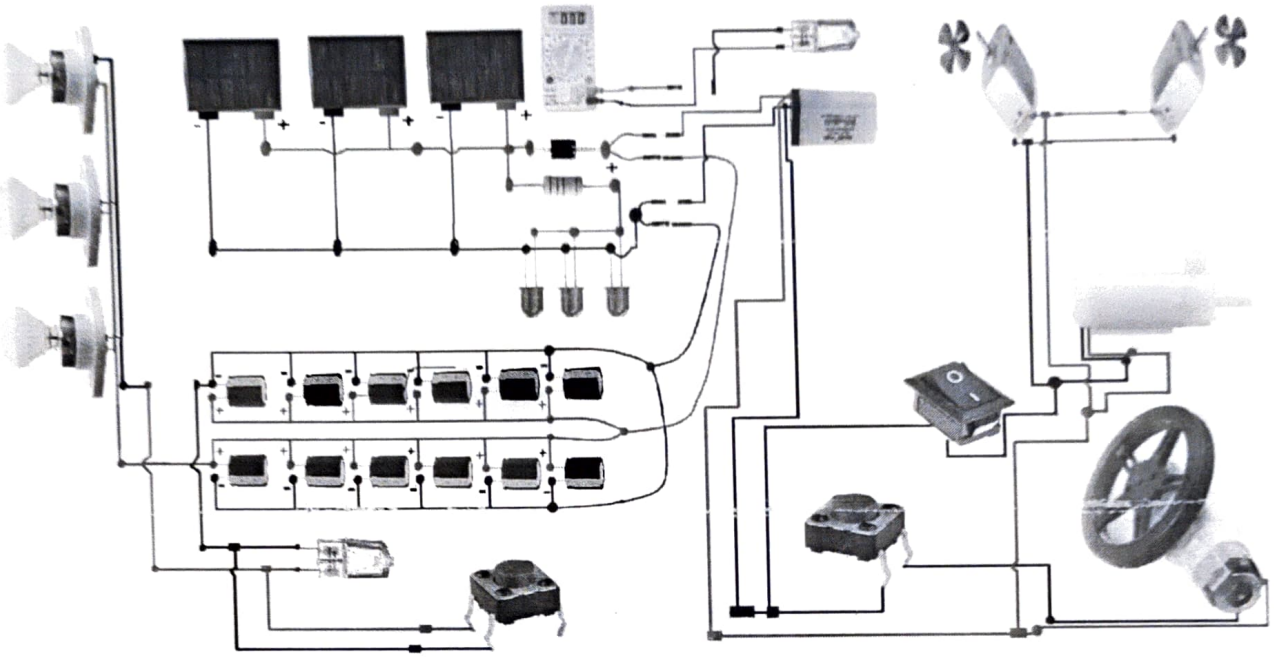
This Project Making Ideas and Invent is My before I make this idea no one make this idea so this project data not available on internet , your idea full related to solar energy and In 1939, Bell Laboratories engineer, Russell Ohl, submitted a patent for what we now know as the modern solar panel. His patent called for the purification of metalloid crystals to form a solar cell. This process created a material that was much more conductive for solar energy and I used solar penal to heating solar penal base so our idea same change . Our Full Project Work on generate electricity by waste material power . So My Full Project Depend on Heating Solar Penal and i am only one who is work on heating solar penal so my idea research data not available on internet .

Basic Working Of Block Diagram:

In This Block Diagram you can see when we burn waste materials and fire box then heat generating and heating panel starts to heat convert electricity and after that that electricity we can see by LED Bulb glowing and that electricity go to circuit and after that in battery and start storing power and when electricity store in battery then heating sensor turn on the output power supply and LED Bulb start glowing and smoke go tc water tank and filter system start controlling pollution.

CHAPTER 2
CIRCUIT DIAGRAM

Circuit Diagram:



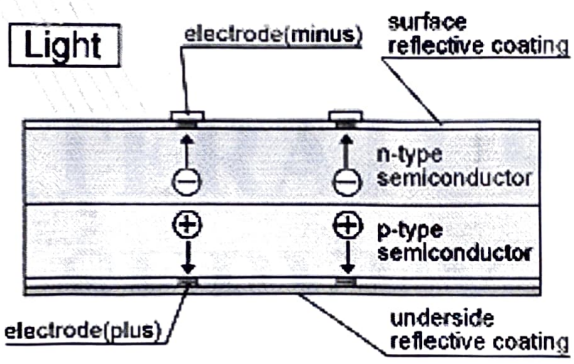
CHAPTER-3
WORKING

Working Principle Of Project:

This Project Working Depend On Heating Solar Penal So Heating Solar Penal Principle is Project Working Principle

Simply put, a Heating panel works by allowing photons, or particles of light or heat , to knock electrons free from atoms, generating a flow of electricity. Heating panels actually comprise many, smaller units called photovoltaic cells. (Photovoltaic simply means they convert heating or light into electricity.)

How to Work Heating Penal



A p-n junction is formed by placing p-type and n-type semiconductors next to one another. The p-type, with one less electron, attracts the surplus electron from the n-type to stabilize itself. Thus the electricity is displaced and generates a flow of electrons, otherwise known as electricity.

When heat hits the semiconductor, an electron springs up and is attracted toward the n-type semiconductor. This causes more negatives in the n-type semiconductors and more positives in the p-type, thus generating a higher flow of electricity. This is the photovoltaic effect.

CHAPTER 4
LITERATURE SURVEY

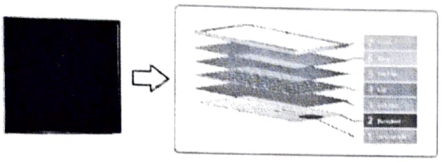
Literature Survey:

Now a day we have more idea for generate electricity , but in your idea we show very different ideas , because this idea we invent before i make nobody make this idea , every day everyone for making food burn many matters so we make a different type of zaar box , when some one making for food start burning that time our zaar box waste extra heat uses for convert heat to electricity and that electricity we can use in real life because we can store that electricity .

Hardware Specifications :- Main Components Working Name

SN	Component
1	Heating Penal
2	Heating Sensor
3	Capacitor 25v/1000uf
4	LED Bulb
5	Resistor
6	DC motor 3000 RPM
7	Battery 4.5V
8	PCB
9	Jack System
10	Switch
11	IN4007
12	Wire

5v Heating Penal



Heating Penal Principle

Voltage -6V

Simply put, a Heating panel works by allowing photons, or particles of light or heat , to knock electrons free from atoms, generating a flow of electricity. Heating panels actually comprise many, smaller units called photovoltaic cells. (Photovoltaic simply means they convert heating or light into electricity.)

An electric heater is an electrical device that converts an electric current into heat. The heating element inside every electric heater is an electrical resistor, and works on the principle of Joule heating: an electric current passing through a resistor will convert that electrical energy into heat energy.

Solar water heating systems use panels or tubes, called solar collectors, to gather solar energy. The solar collectors convert the infra-red portion of visible light into heat. They are filled with a mix of water and glycol. This fluid is pumped round a circuit, which passes through the hot water cylinder.

Heating Sensor/ Tubelight Starter



This sensor gives the battery power to the LED bulb only when this sensor is heated by heating . If this sensor is not heated , then LED bulbs will not glowing.

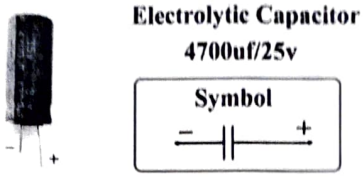
Here is heating penal use for switching battery power , Because A Heating Sensor is an electrically operated switch.

Temperature sensors work by providing readings via electrical signals. Sensors are composed of two metals that generate an electrical voltage or resistance when a temperature

smce.ac.in ESTD : 2006

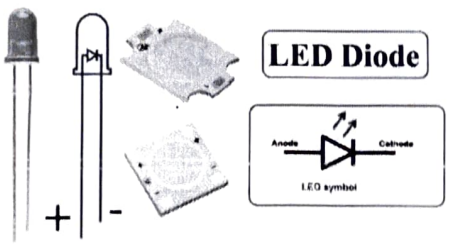
change occurs by measuring the voltage across the diode terminals. When the voltage increases, the temperature also increases.

Starter is a small device used in certain types of lighting, especially fluorescent tubes. It usually consists of a small glass container with argon gas inside and a bimetallic foil. Basically, a fluorescent starter generates a sufficient discharge to "activate" the gas inside the lamp so that it lights up.

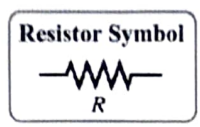


A voltage applied across the conductors creates an electrical field in the capacitor, which stores energy. A capacitor operates like a battery in that, if a potential difference is applied across it that can cause a charge greater than its "present" charge, it will be charged up.

Used to reduce voltage fluctuations in various filtering devices. Used in output and input smoothing to filter when the DC signal is weak with the AC component. They are extensively used for noise filtering or decoupling in power supplies.



- 3V DC LED Bulb We use for showing electricity generating
- EDs work on the principle of Electroluminescence. On passing a current through the diode, minority charge carriers and majority charge carriers recombine at the junction. On recombination, energy is released in the form of photons.



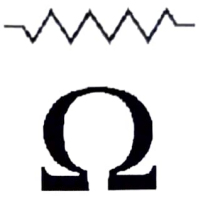
100 Ω Resistor

5 Watt





resistor



resistor symbol

- We use resistor with LED bulb
- Commonly used in breadboards and other prototyping applications, these 100 ohm resistors make excellent pull-ups, pull-downs and current limiters. These thick-lead versions of the resistors fit snugly into a breadboard with very little movement, so you should have few to no issues using them in your next project!

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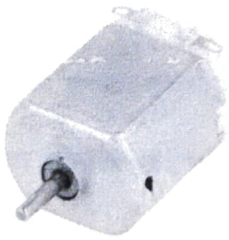
DC Motor

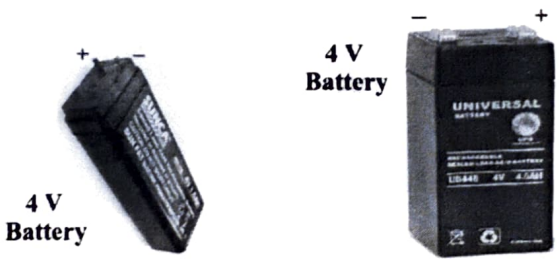
RPM: 3000, Operating Voltage: 6V

DC,

Shaft diameter: 6mm with internal hole, Torque: 7 kg-cm

No-load current = 60 mA(Max), Load current = 300 mA(Max).





(This is rechargeable battery ,Charging For Contact 4V Charger And Battery Charge)

(4V Battery More Use For LED Bulb , Mini Circuit and Mini Motor)

Battery is a device that stores energy and then discharges it by converting chemical energy into electricity. Typical batteries most often produce electricity by chemical means through the u of one or more electrochemical cells.

https://energyeducation.ca › Battery

Battery - Energy Education

- We use resistor with LED bulb

DC Water Pump Motor

RPM: 3000, Operating Voltage: 6V

DC,

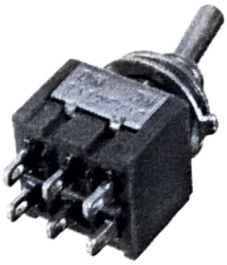
Shaft diameter: 6mm with internal hole, Torque: 7 kg-cm

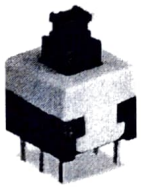
No-load current = 60 mA(Max), Load current = 300 mA(Max).



DC water pump is a small electric water pump motor that powered by a 24v, 12v, 5v, 6v, or 3v direct current power supply, solar panel or battery. It use centrifugal force to pressurize, trsanfer or circulating water or other liquids.

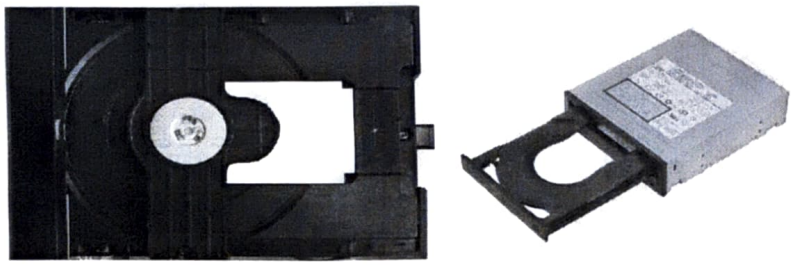
6 PIN ON/OFF Switch





6 Pin Push Switch also known as Mini DPDT Push Switch, is nothing but a combination of two push switches placed together inside one package. Unlike momentary switches which connect the wires of the switch only for a second, this switch retains its ON-OFF state till pushed later on.

DVD Gate (DVD Trolley)



(More use in open close/ Home Door and Gate System For Project)

CHAPTER 5
ADVANTAGES

Advantages:

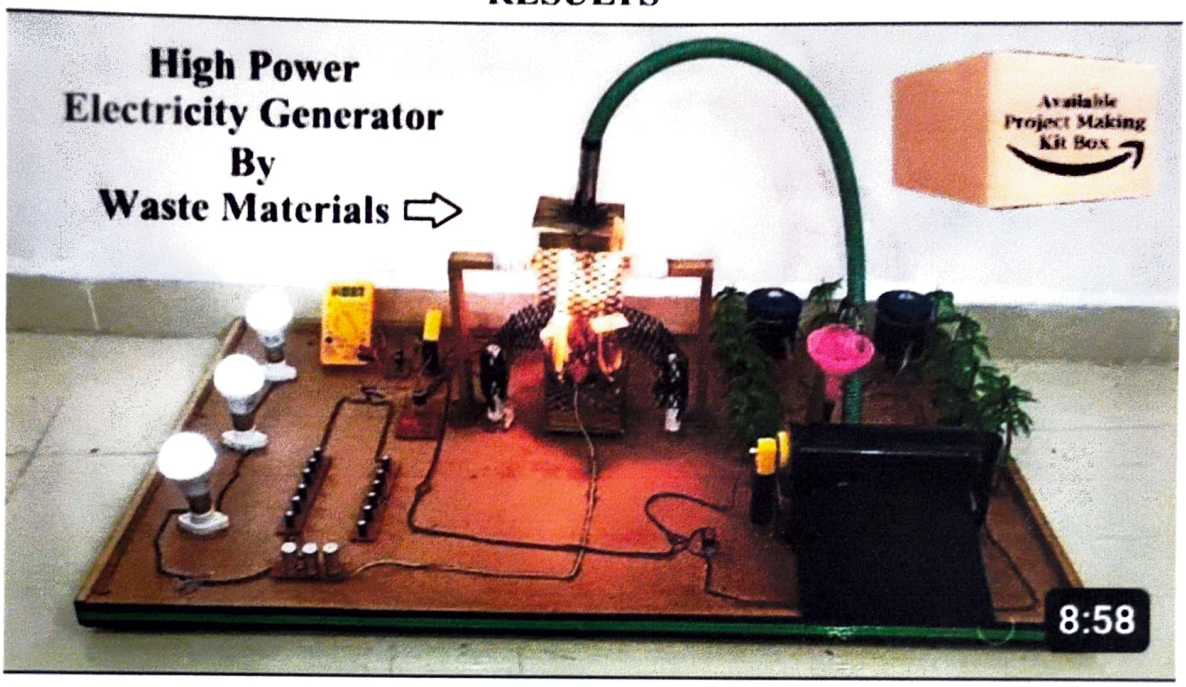
- He can generate electricity by solid waste
- He can generate electricity within second and anywhere
- We can use any where
- Making cost very less and any where we can make
- He can generate electricity depend on heating penal voltage and zaar box size
- We can generate electricity making food time.
- We can collect Corban by this idea and make many product by Corban

Limitations of the system:

1. We can not burn waste materials in large level so we can generate electricity only normal level .
2. We can control pollution 100% when we burn plastic and other .

CHAPTER 6
RESULTS

RESULTS


High Power
Electricity Generator
By
Waste Materials
Available
Project Making
Kit Box
8:58

CHAPTER 7
CONCLUSION

CONCLUSION:

When We Completed our project after that we check our project working , it is very good ,he is full work without stop or any problem , So We can use our Project in real life like our industries or Field and he perfect work on there without any problem , This is our project idea and it is full work our project . So Our project idea is work in our prototype project ,so our conclusion is so good.

Future development

- Effectiveness of this project can be improved by following this
- Recommendation:
- We can make high quality heating penal for generate high electricity
- We can make large level burning box with easily heating penal connecting system
- We can make best storage system by generate electricity by waste materials

CHAPTER 8
REFERENCE

Reference

1) Literature Survey of **Luminous, Su-Kam** from google. com
2) Literature Survey of Md. A. Hussainfrom google. Com
3) https://www.flipkart.com
4) https://www.electronicscomp.com
5) www.googleimages.com
6) www.reserchgate.com
7) Various web pages form google for search of Capstone project